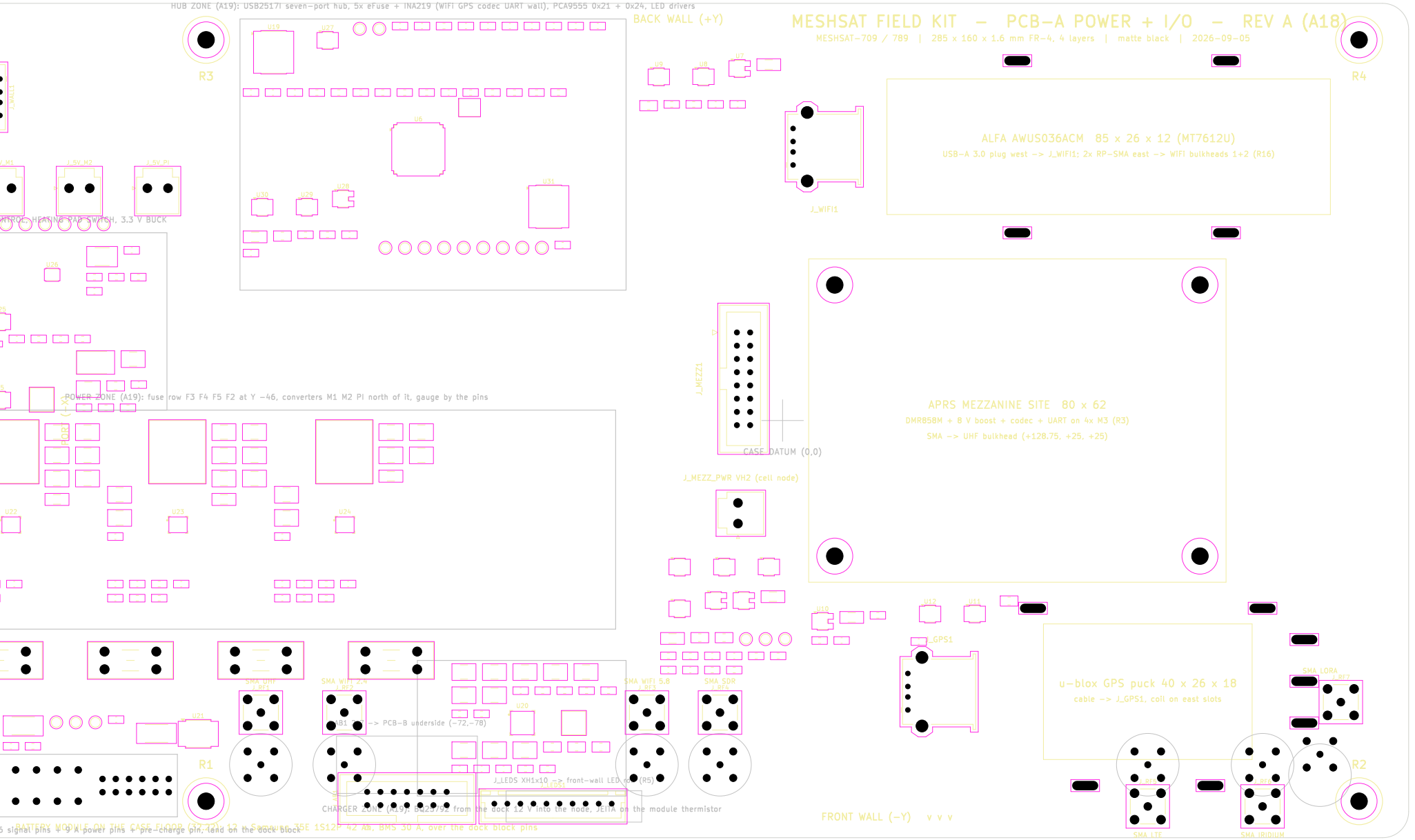
MESHSAT FIELD KIT — PCB-A POWER + I/O — REV A (A18)
MESHSAT-709 / 789 | 285 x 160 x 1.6 mm FR-4, 4 layers | matte black | 2026-09-05
HUB ZONE (A19): USB2517I seven-port hub, 5x eFuse + INA219 (WIFI GPS codec UART wall), PCA9555 0x21 + 0x24, LED drivers
BACK WALL (+Y)
R3
R4
ALFA AWUS036ACM 85 x 26 x 12 (MT7612U)
USB-A 3.0 plug west -> J_WIFI1; 2x RP-SMA east -> WIFI bulkheads 1+2 (R16)
J_WIFI1
MIRROR HEATING PAD SWITCH, 3.3 V BUCK
POWER ZONE (A19): fuse row F3 F4 F5 F2 at Y -46, converters M1 M2 PI north of it, gauge by the pins
PORT (+X)
APRS MEZZANINE SITE 80 x 62
DMR858M + 8 V boost + codec + UART on 4x M3 (R3)
SMA -> UHF bulkhead (+128.75, +25, +25)
J_MEZZ1
CASE DATUM (0,0)
J_MEZZ_PWR VH2 (cell node)
u-blox GPS puck 40 x 26 x 18
cable -> J_GPS1, coil on east slots
J_GPS1
SMA UHF
SMA WIFI 2.4
SMA WIFI 5.8
SMA SDR
SMA LORA
SMA LTE
SMA IRIDIUM
R1
R2
J_LEDS XH1x10 -> front-wall LED ring (R5)
CHARGER ZONE (A19): BQ25792 from the dock 12 V into the NODE, JEITA on the module thermistor
FRONT WALL (-Y) v v v
STARBOARD (+X)
6 signal pins + 3 A power pins + pre-charge pin, land on the dock block